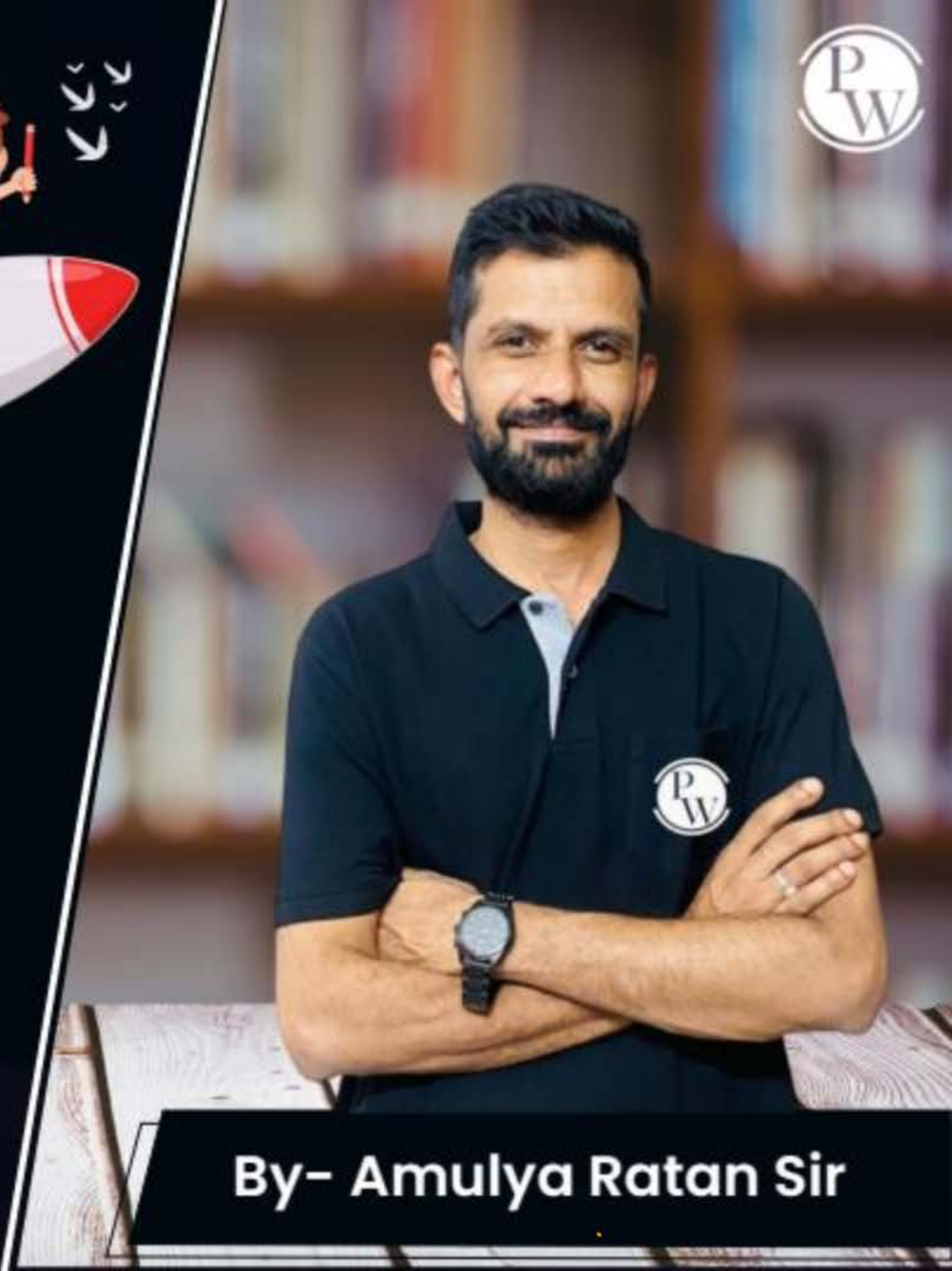
GATE ALL BRANCHES



General Aptitude

ANALYTICAL APTITUDE

Lecture No.- 05



By- Amulya Ratan Sir

Recap of Previous Lecture



Topic Problem Solving

Topics to be Covered



Topic-1 Cubes

Topic-2 Dices ?

[MCQ]

#Q. Five persons P, Q, R, S and T belongs to five different cities Jabalpur, Bhopal, Hyderabad, Raipur and Delhi. They are invited to deliver a seminar on different subject Viz., Chemistry, Physics, Mathematics, History and Economics but not necessarily in the same order.

P who belongs to Bhopal gives seminar on Economics. The person who gives seminar on Physics belongs to Jabalpur. T does not give seminar on Chemistry or History but belongs to Hyderabad. S does not belongs to Raipur but gives seminar on Chemistry and History is the option of R.



Q1. Who belongs to Raipur? → R

Q2. Physics seminar will be given by? → Q

Q3. T gives seminar on? → Maths

Q4. S belongs to which city? → Delhi

Q5. Who belongs to Jabalpur? → Q

- Five persons P, Q, R, S and T belongs to five different cities Jabalpur, Bhopal, Hyderabad, Raipur and Delhi. They are invited to deliver a seminar on different subject Viz., Chemistry, Physics, Mathematics, History and Economics but not necessarily in the same order.
- P who belongs to Bhopal gives seminar on Economics.
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- S does not belongs to Raipur but gives seminar on Chemistry and History is the option of R.

	CITY	SUBJECT
P	Bhopal	Economics
Q	Jabalpur	Physics
R	Raipur	Hist
S	Delhi	Chem
T	Hyderabad	Maths

CITY	JABALPUR	BHOPAL	HYDERABAD	RAIPUR	DELHI
SUBJECT	Chem	Phy	Maths	Hist	Eco

[MCQ]

#Q. On a shelf are placed six volumes side by side labeled A, B, C, D, E and F. Three volumes B, C and E have green covers while the other volumes have yellow covers. A, D and B are new volumes while the rest are old volumes. A, C and B are law reports, while the rest are Gazetteers.



[MCQ]

#Q. Which volume is new, yellow covered and a Gazetteer?

A E

B D

C C

D A

[MCQ]

#Q. Which two volumes are old Gazetteers?

A. E and F

B. D and B

C. C

D. F

[MCQ]

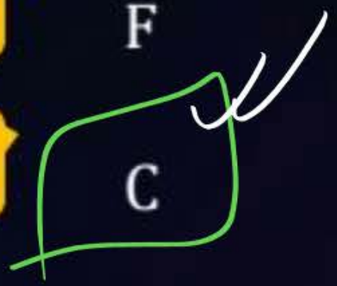
#Q. Which is green covered, old and a law report volume?

A E

B D

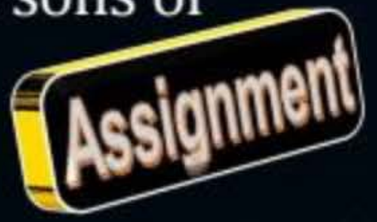
C F

D C



[MCQ]

#Q. Prashant Arora has three children – Sangeeta, Vimal and Ashish. Ashish married Monika, the eldest daughter of Mr. and Mrs. Roy. The Roys married their youngest daughter to the eldest son of Mr. and Mrs. Sharma, and they had two children name Amit and Shashi. The Roys have two more children, Roshan and Vandana, both elder to Veena. Sameer and Ajay are sons of Ashish and Monika. Rashmi is the daughter of Amit.



Q1. What is the surname of Rashmi? Sharma

Q2. How is Sameer related to the father of Monika? Grand son

Q3. What is the surname of Sameer? Arora

Q4. How is Mrs. Roy related to Ashish? Mother-in-law

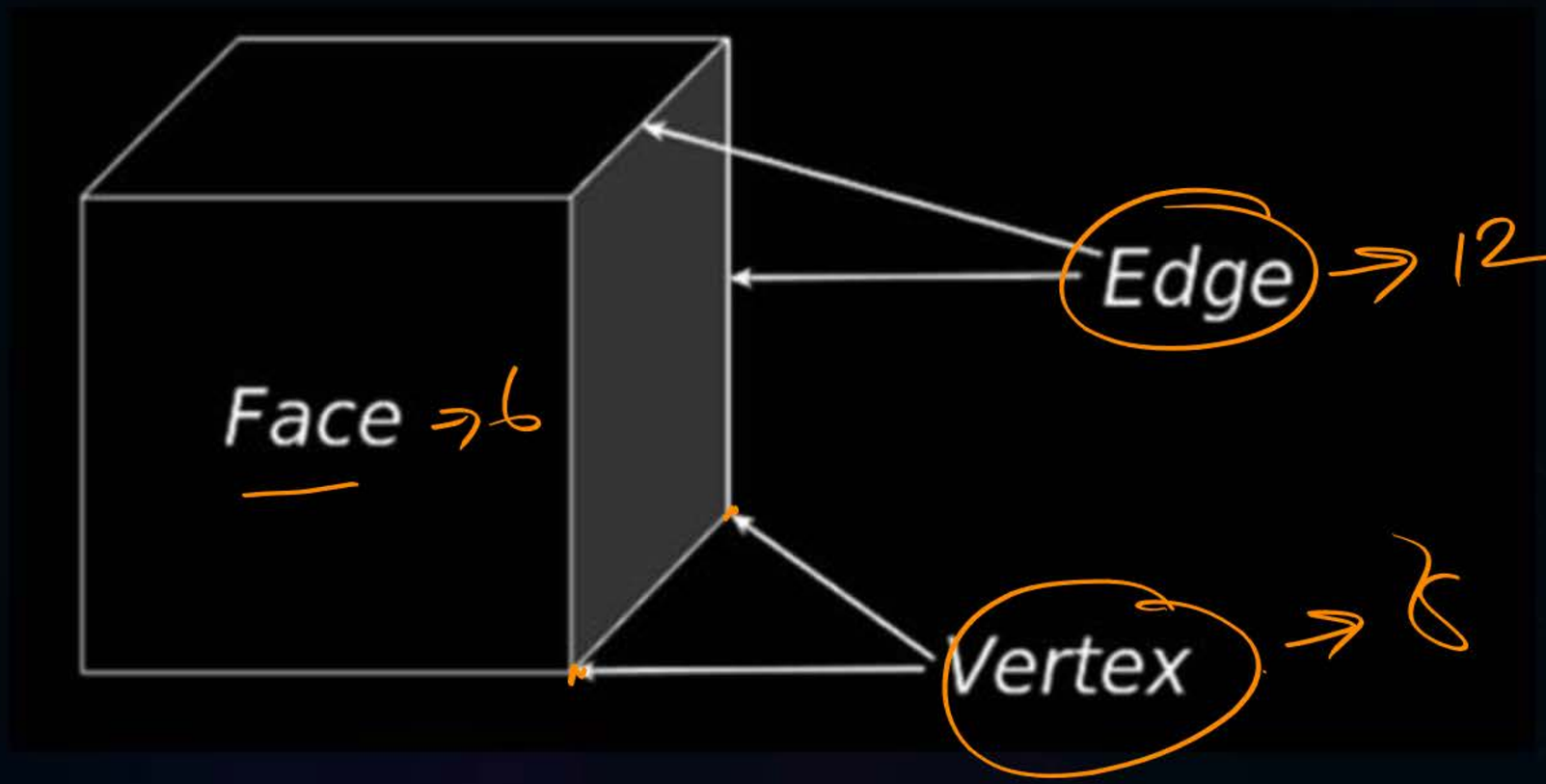
Cubes

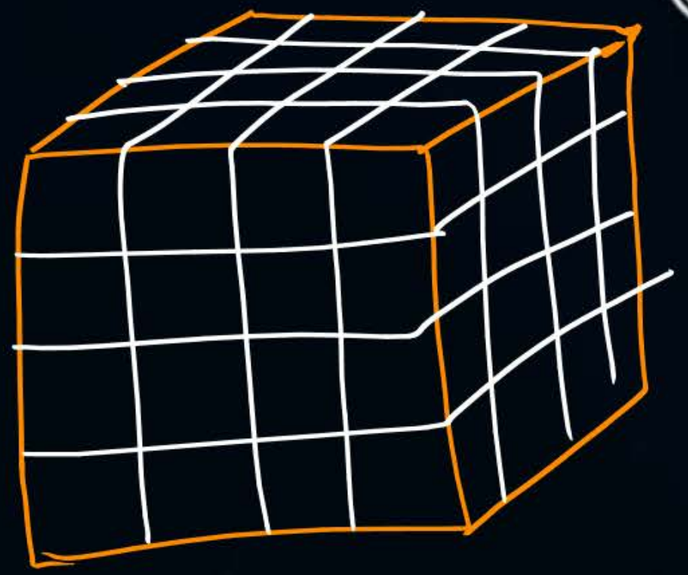


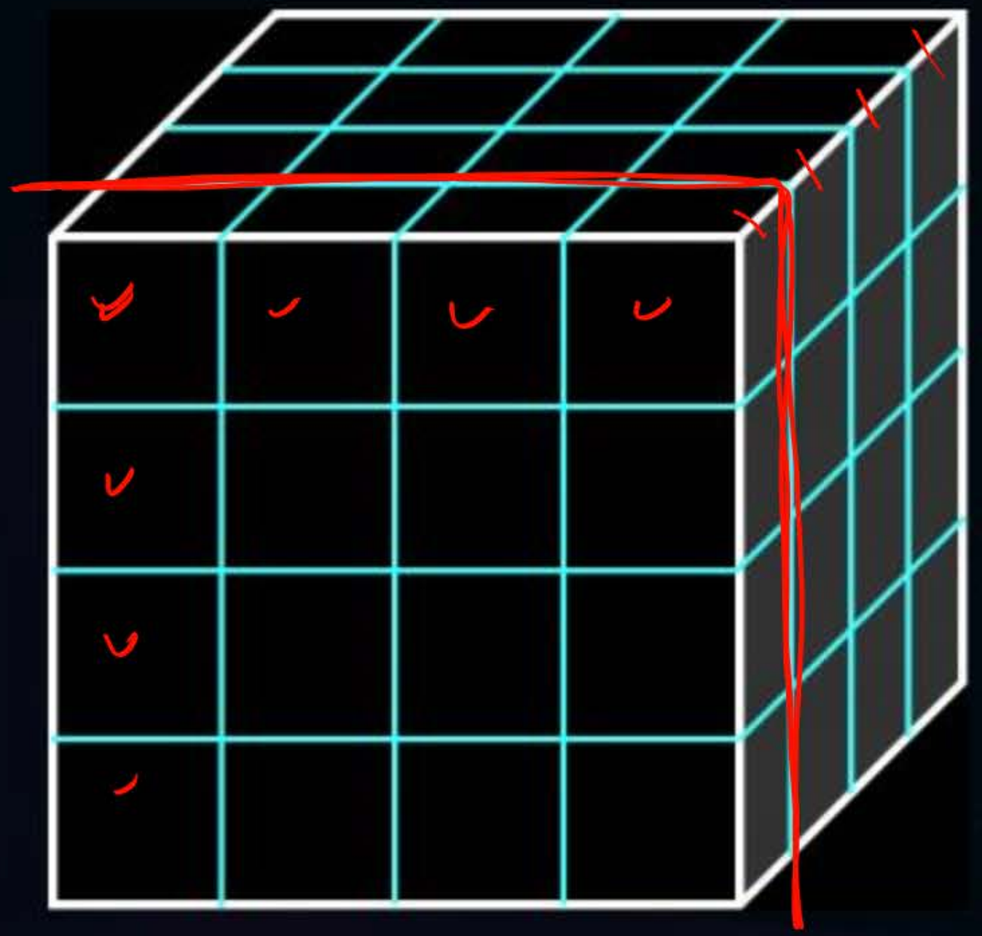
faces → 6

Edges → 12

Vertex → 8


Edge → 12
Face → 6
Vertex → 8





$16 \times 4 = 64$

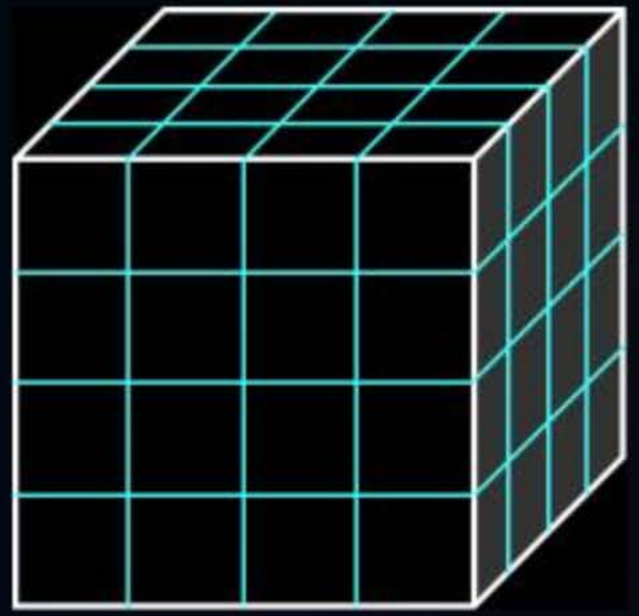
3
'x' cut

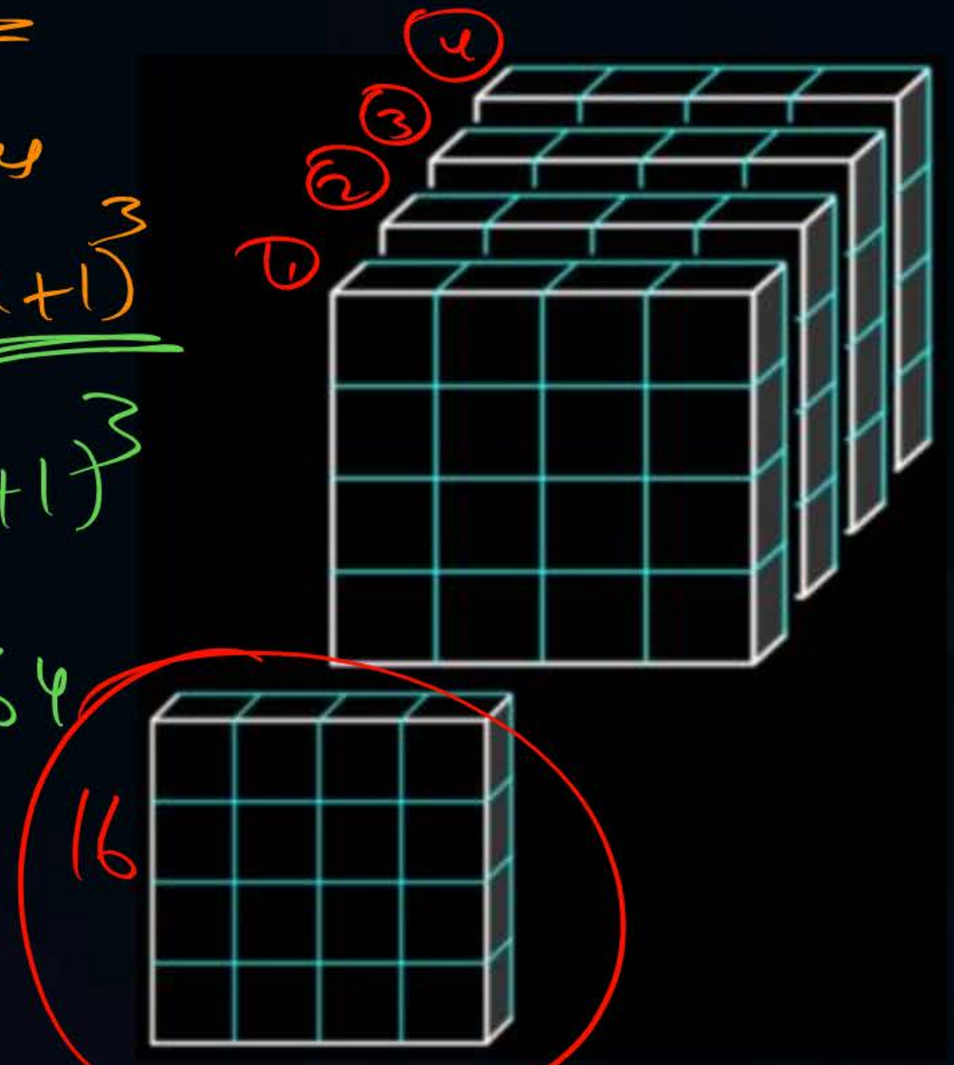
No. of cubes

$= (x+1)^3$

$(3+1)^3$

$4^3 = 64$





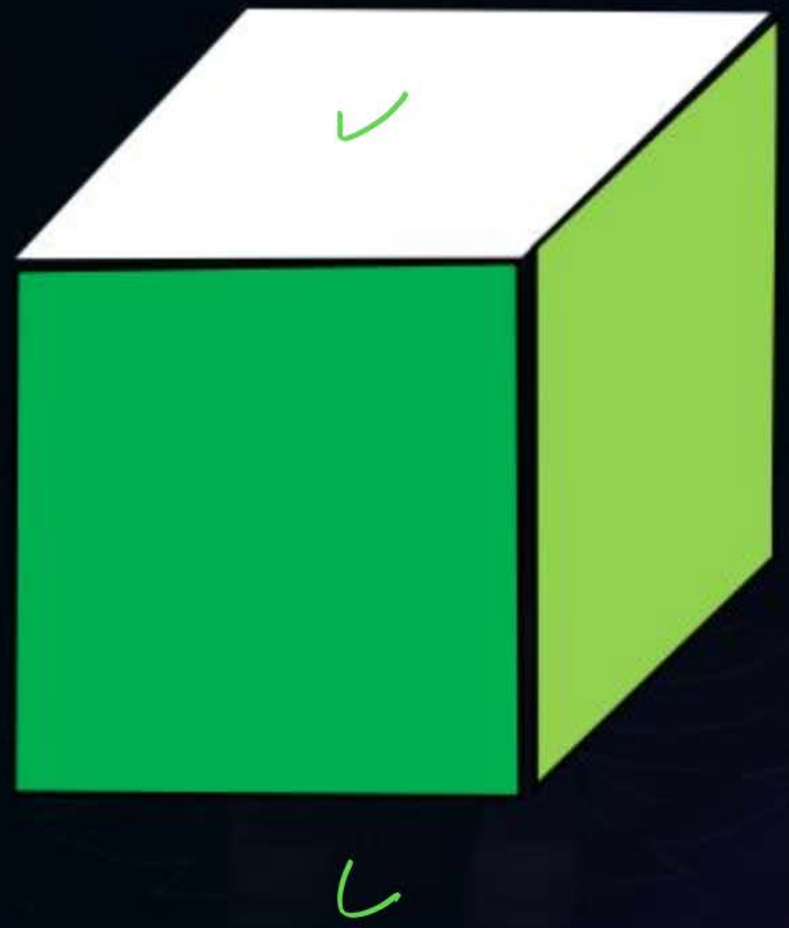
① — ⑦

[MCQ]

#Q. A Cube was painted on one of the opposite pair faces as WHITE & remaining four faces as GREEN. Five cuts were given on each edges to form smaller Cubes of equal size.

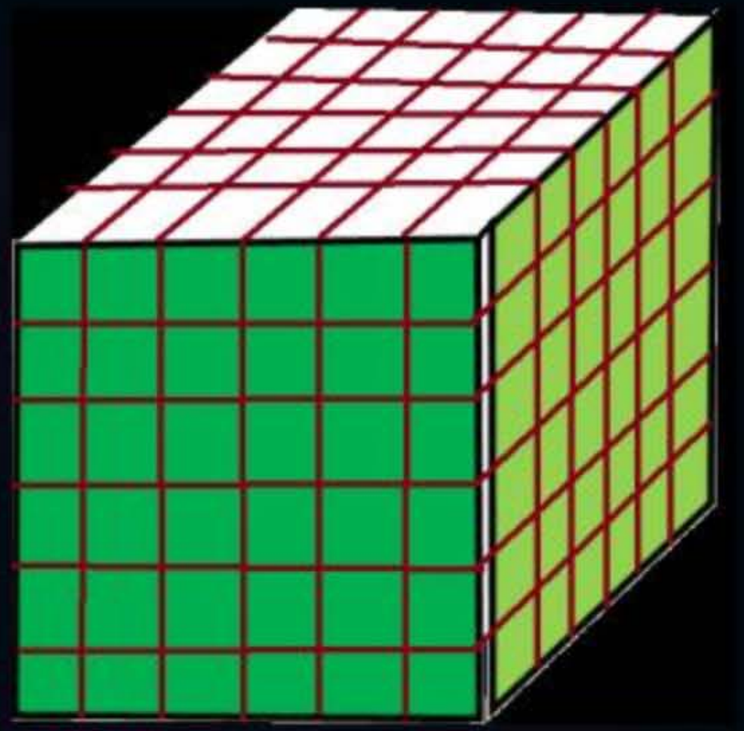
1. How many cubes will be formed?
2. How many cubes will have no face painted at all?
3. How many cubes will have only one face painted?
4. How many cubes will have only two face painted, but with same colour?
5. How many cubes will have only two face painted, but different colour?
6. How many cubes will have atleast two face painted?
7. How many cubes will have three face painted?

✓A Cube was painted on one of the opposite pair faces as white & remaining four faces as green. Five cuts were given on each edges to form smaller Cubes of equal size.



A Cube was painted on one of the opposite pair faces as white & remaining four faces as green. Five cuts were given on each edges to form smaller Cubes of equal size.

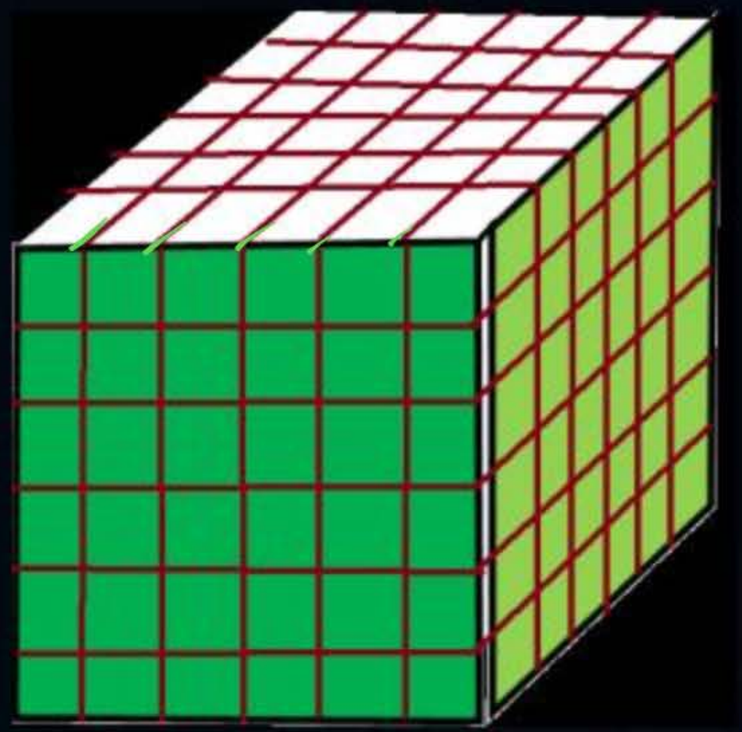




[MCQ]

#Q. How many cubes will be formed?

$(5+1)^3 = 6^3 = 216$



[MCQ]

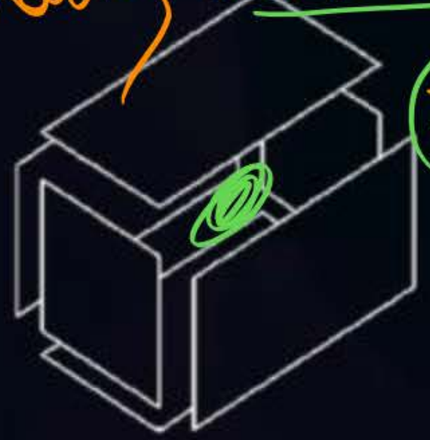
#Q. How many cubes will have no face painted at all?

'5' 'x' cut ⇒ $6^3 = 216$

No. of cubes = $(x+1)^3$

No. of cubes having no face painted

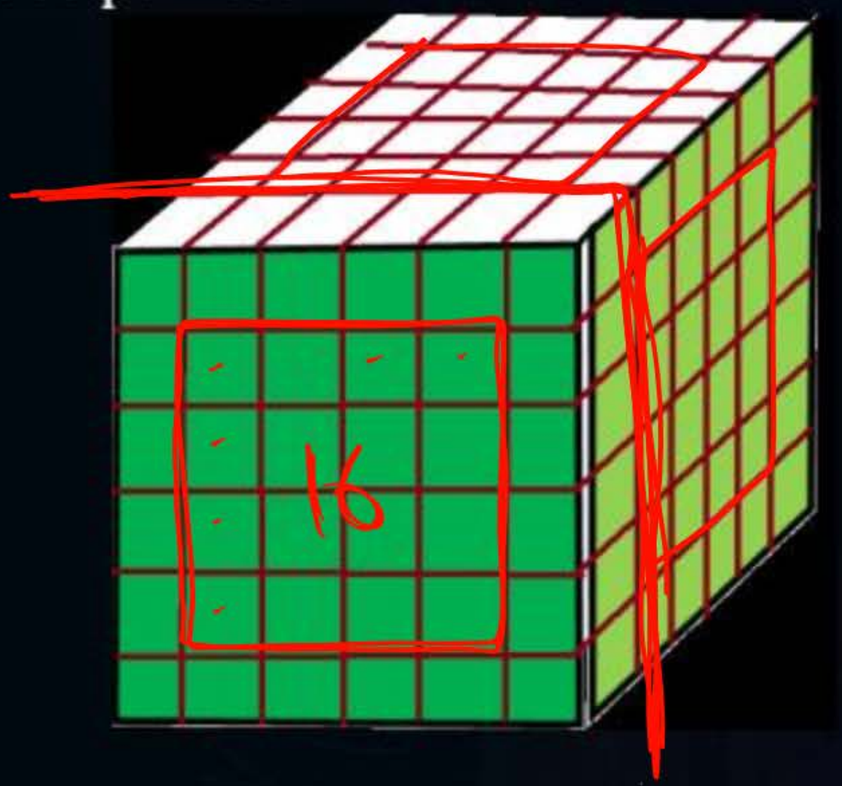
$= (x-1)^3$



$(5-1)^3 = 4^3 = 16 \times 4 = 64$

[MCQ]

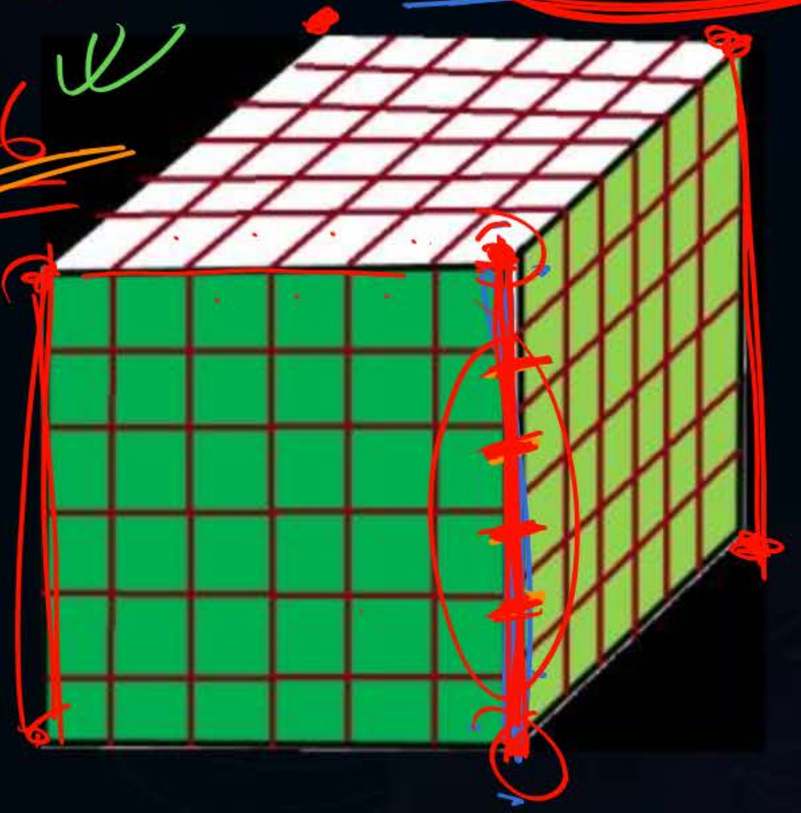
#Q. How many cubes will have only one face painted?



$6 \times 16 = 96$

[MCQ]

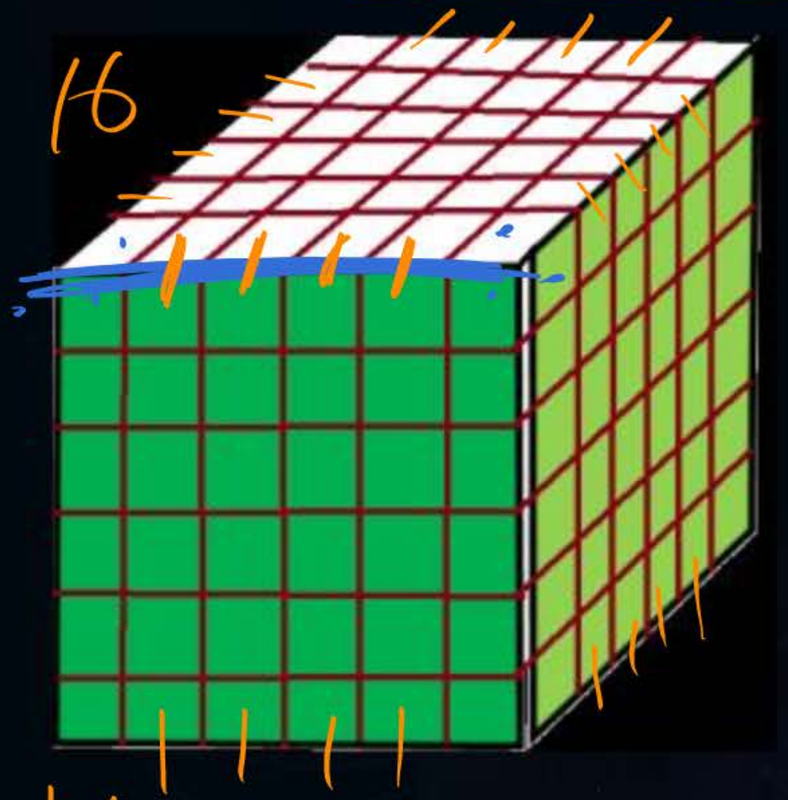
#Q. How many cubes will have only two face painted, but with same colour?



[MCQ]

#Q. How many cubes will have only two face painted, but with different colour?

$16 + 16 = 32$

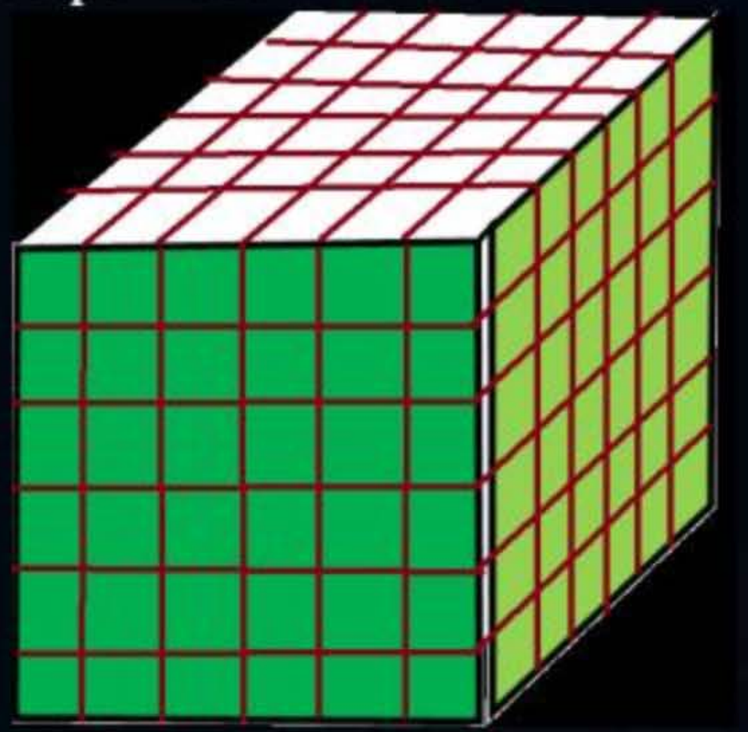


Q How many cubes will have only two faces painted?

$16 + 32 = 12 \times 4 = 48$

[MCQ]

#Q. How many cubes will have atleast two face painted?



Only two face = 48

Three face = 8

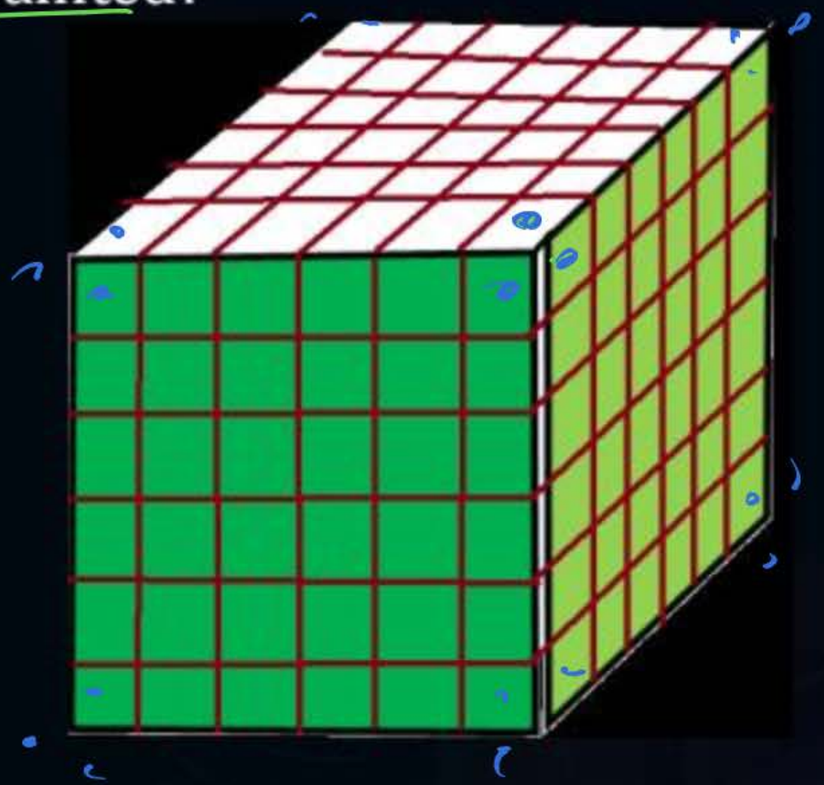
48 + 8

= 56

[MCQ]

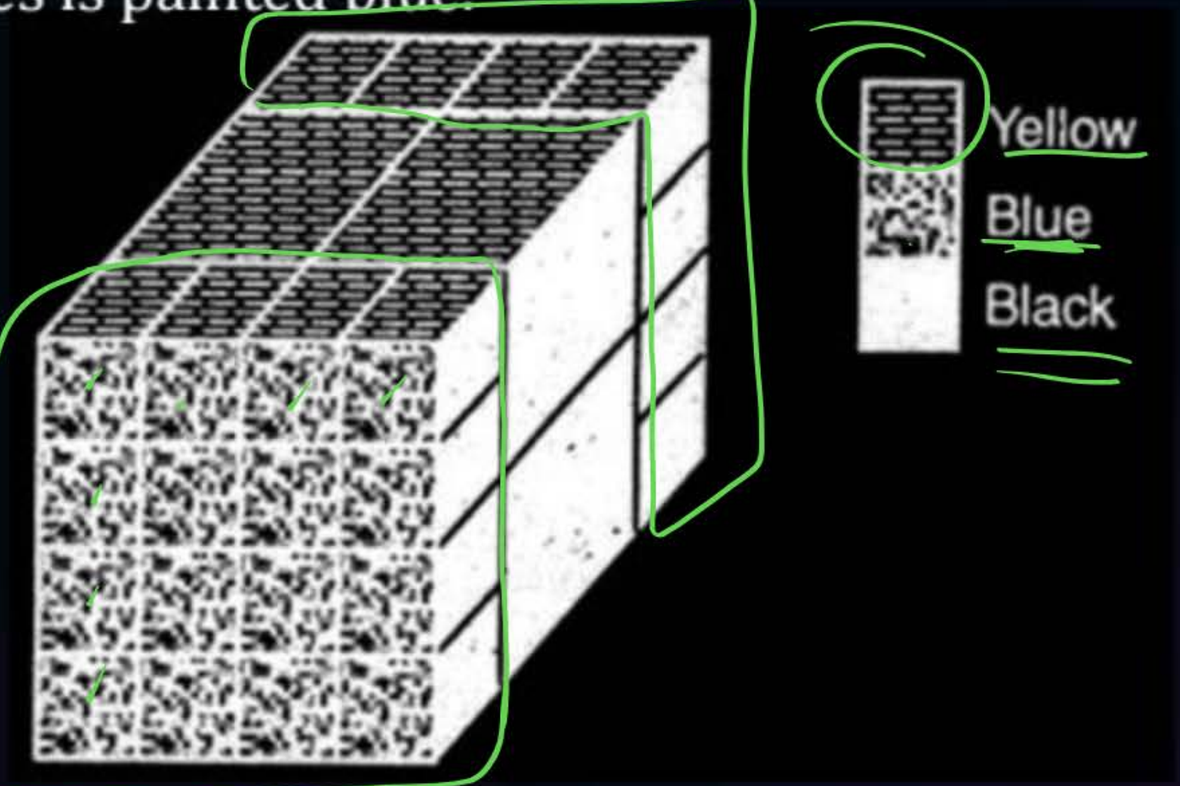
#Q. How many cubes will have three face painted?

8



[MCQ]

#Q. A solid cube as shown in below image, has been painted yellow, blue and black on pairs of opposite face. The cube is the cut into 36 smaller cubes such that 32 cubes are of the same size while 4 others are of bigger size. Also no face of any of the bigger cubes is painted blue.

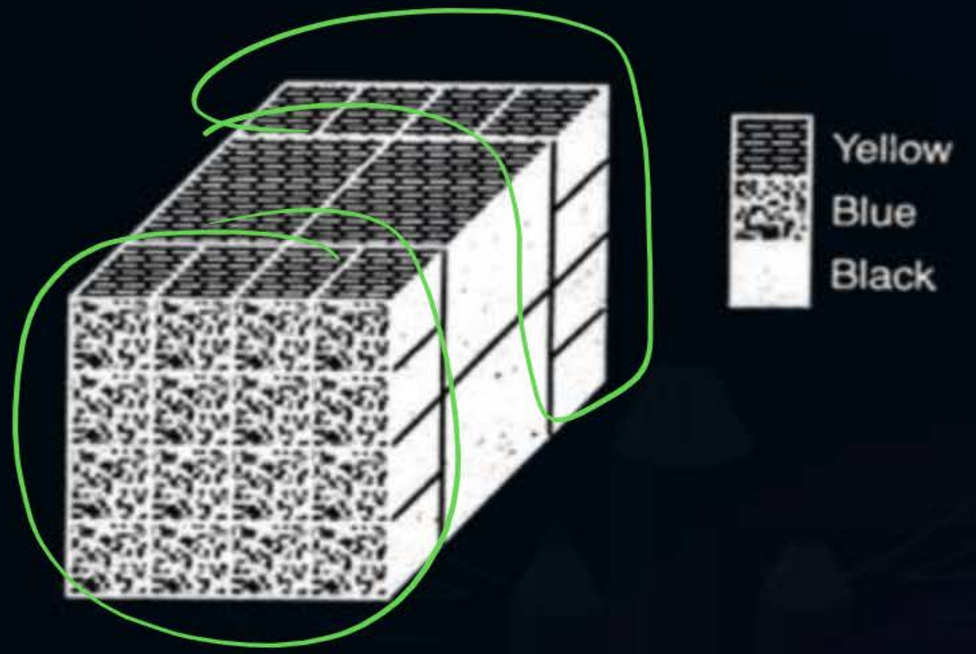


[MCQ]

#Q. How many cubes have at least one face painted blue?

16 + 16

= 32



A 0

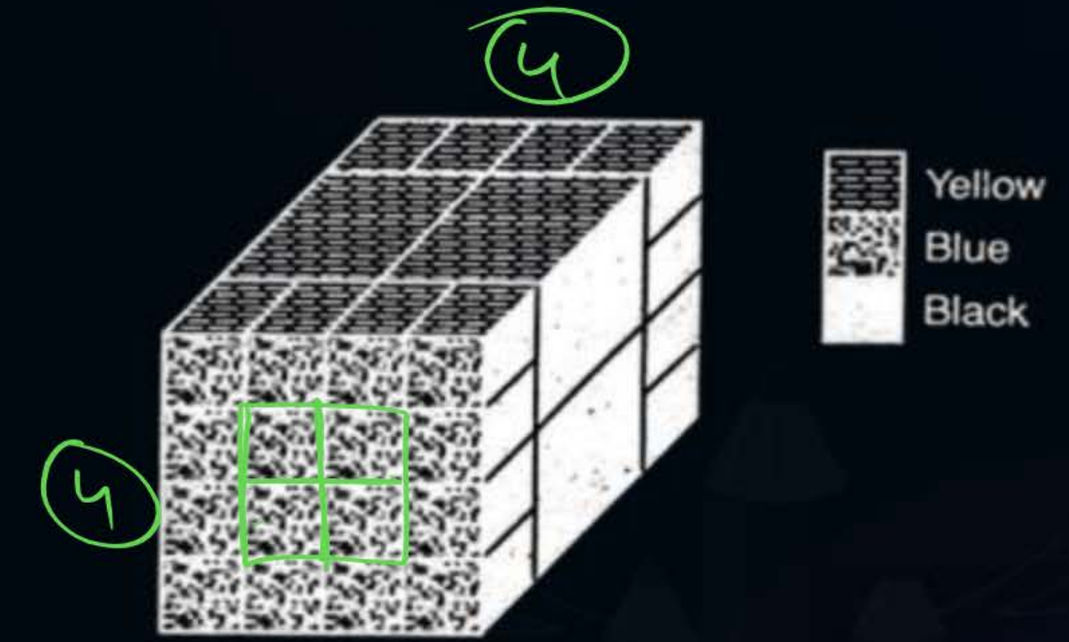
B 8

C 16

D 32

[MCQ]

#Q. How many cubes have only one face painted?



A 12

B 8

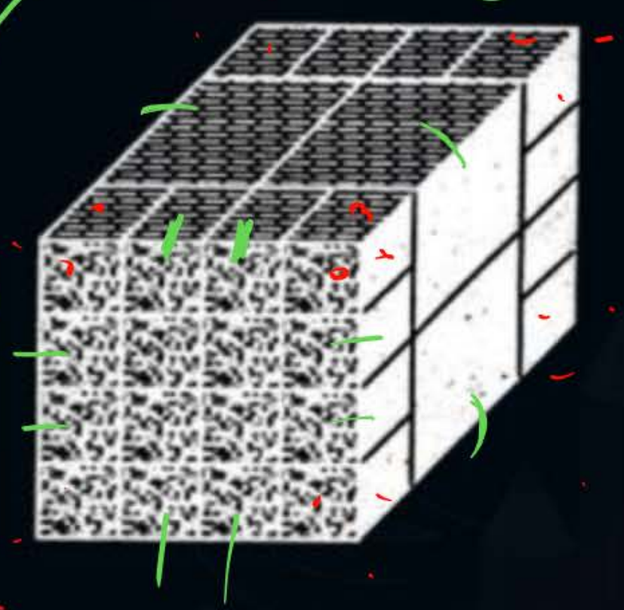
C 4

D 0

4 + 4 = 8

[MCQ]

#Q. How many cubes have two or more faces painted?



A 34

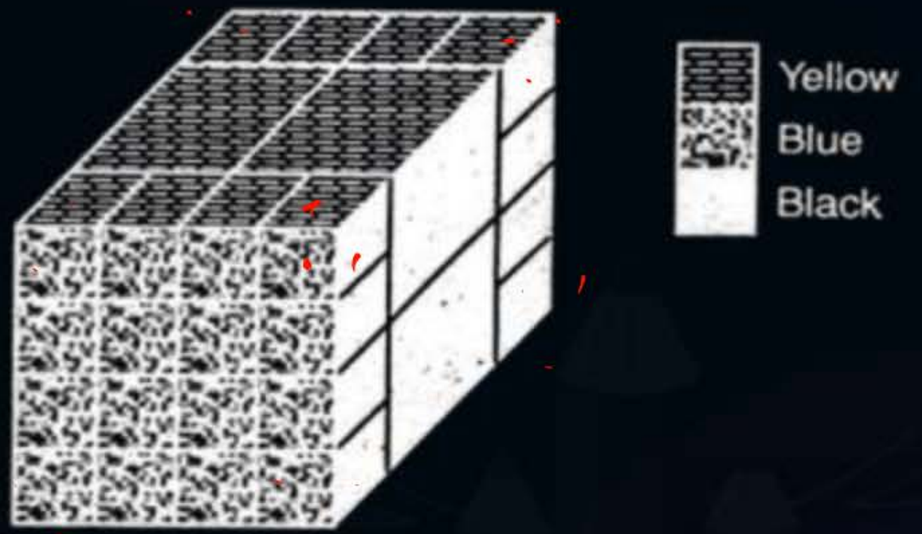
B 36

C 28

D 24

[MCQ]

#Q. How many cubes only three faces painted?



A 0

B 2

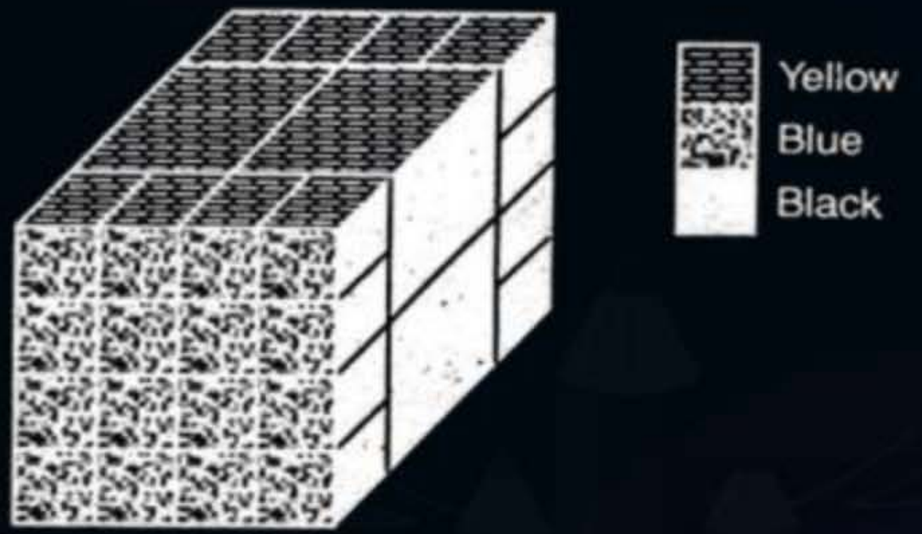
C 4

D 8

[MCQ]

#Q. How many cubes do not have any of their faces painted yellow?

Assignment



A 0

B 8

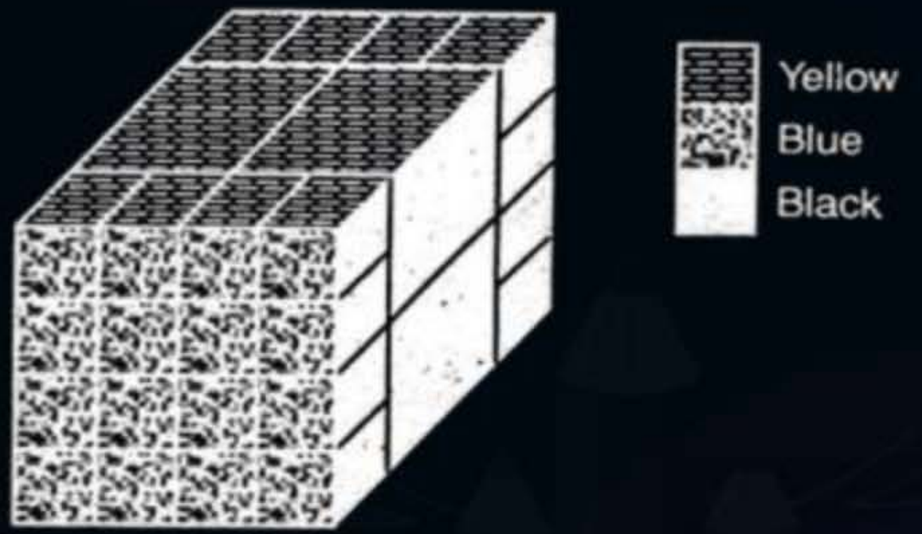
C 16

D 20

[MCQ]

#Q. How many cubes have at least one of their faces painted black ?

Assignment



A 32

B 16

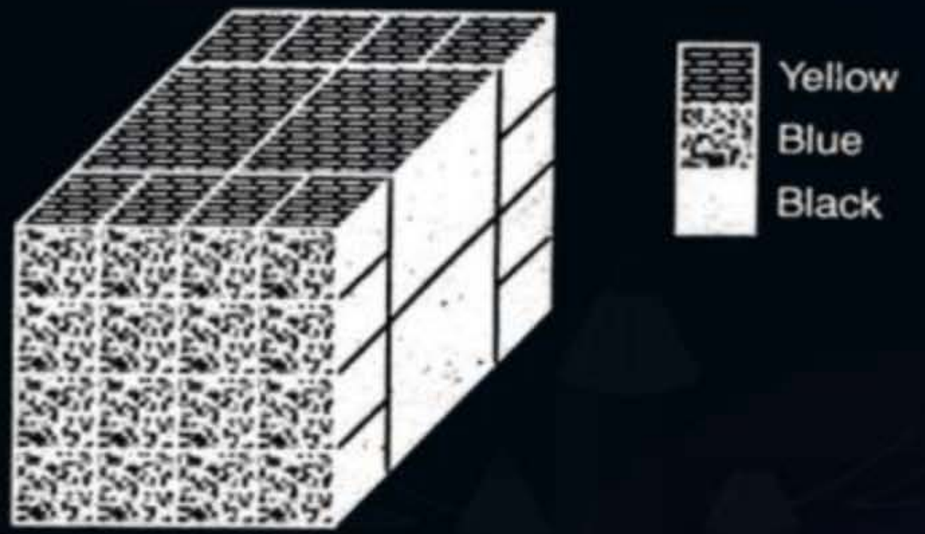
C 8

D 0

[MCQ]

#Q. How many cubes have at least one of their faces painted yellow or blue?

Assignment



A 0

B 16

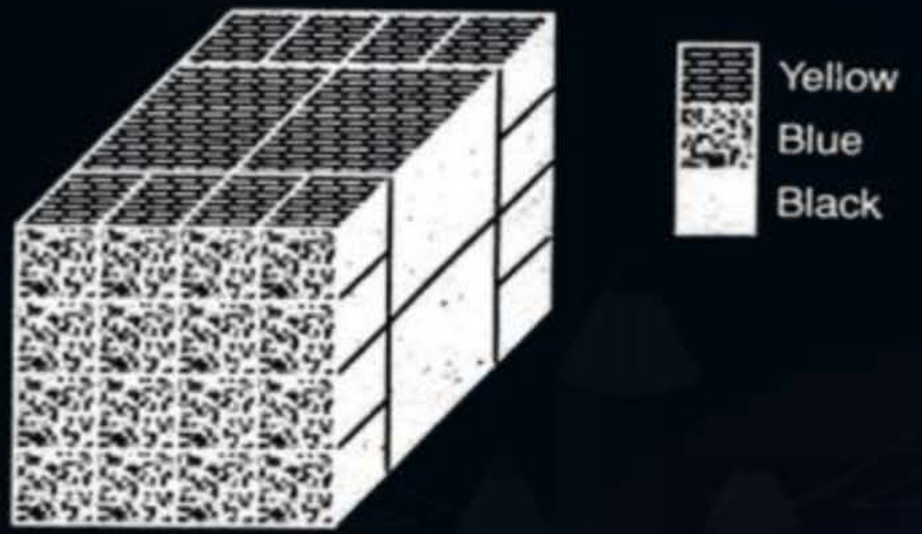
C 32

D 36

[MCQ]

#Q. How many cubes have no face painted?

Assignment



A 0

B 8

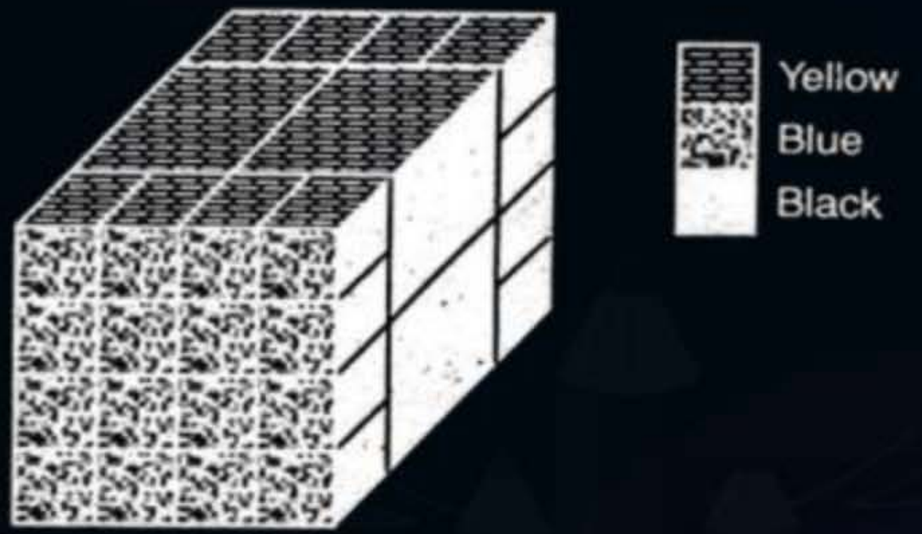
C 12

D 16

[MCQ]

#Q. How many cubes have two faces painted yellow and black respectively?

Assignment



A 16

B 12

C 8

D 0



2 mins Summary

Topic

Cubes & Dice

THANK - YOU